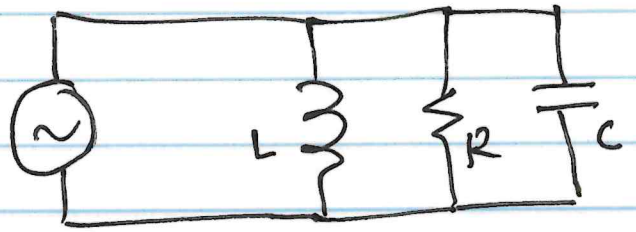
Lecture 7

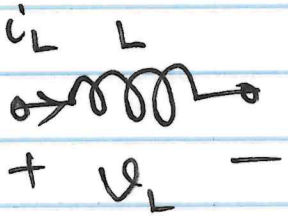


L → * connection between current magnetic field. – Josh.

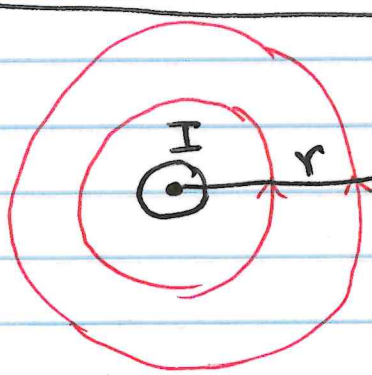
* it resist the change of magnetic field / current – Anav.

* inductors don't like jump in current – Tony.

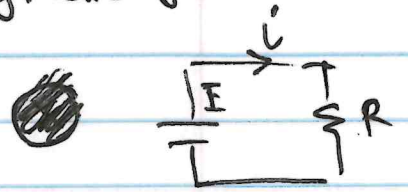
* induce a current in another object – Krishen.



$$v_L = L\frac{di}{dt}$$



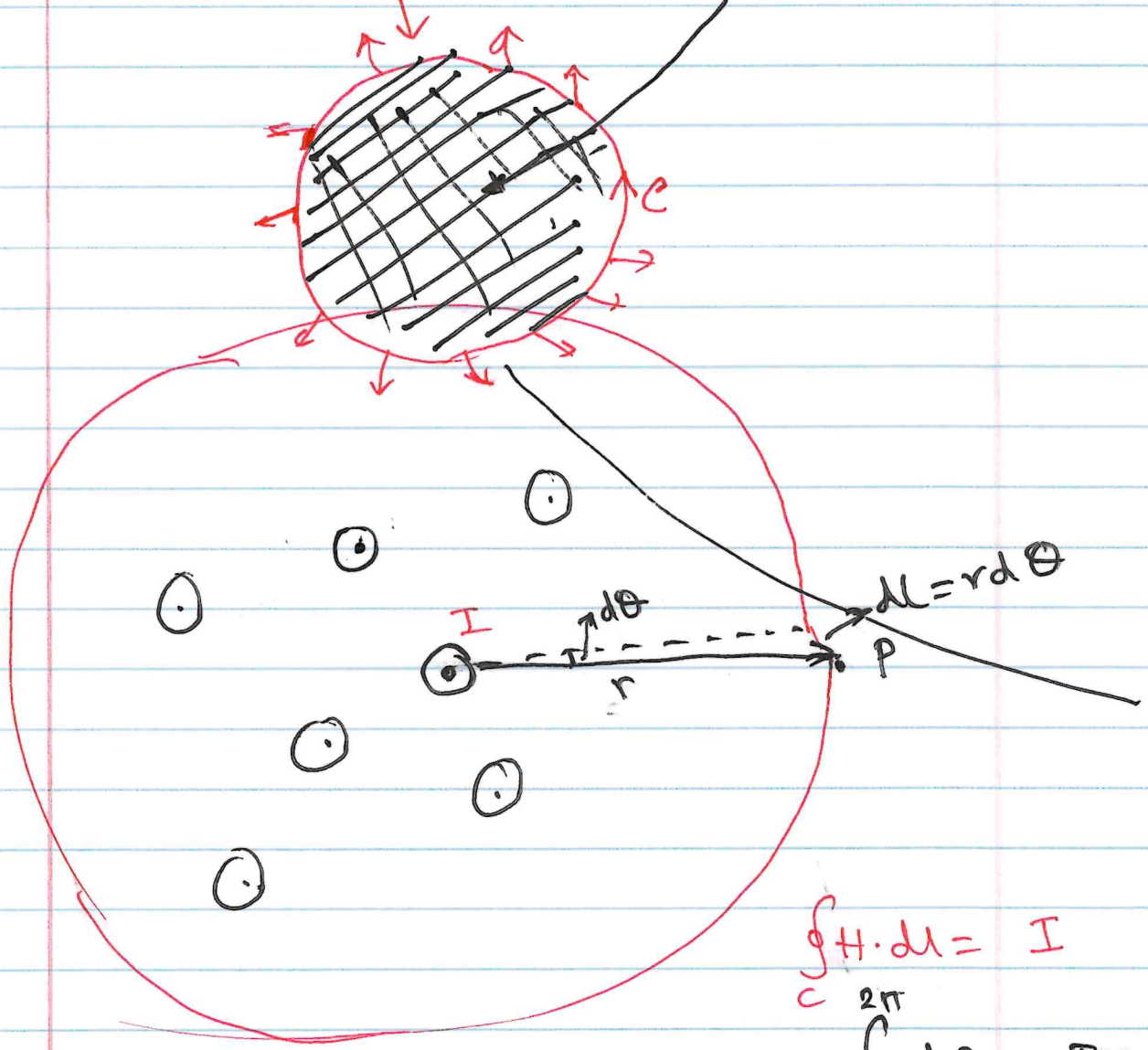
$\bar{H}$, magnetic field A/m.



$$\nabla \times \bar{H} = J + \frac{\partial \bar{D}}{\partial t}$$

Magnetoquasistatic problem

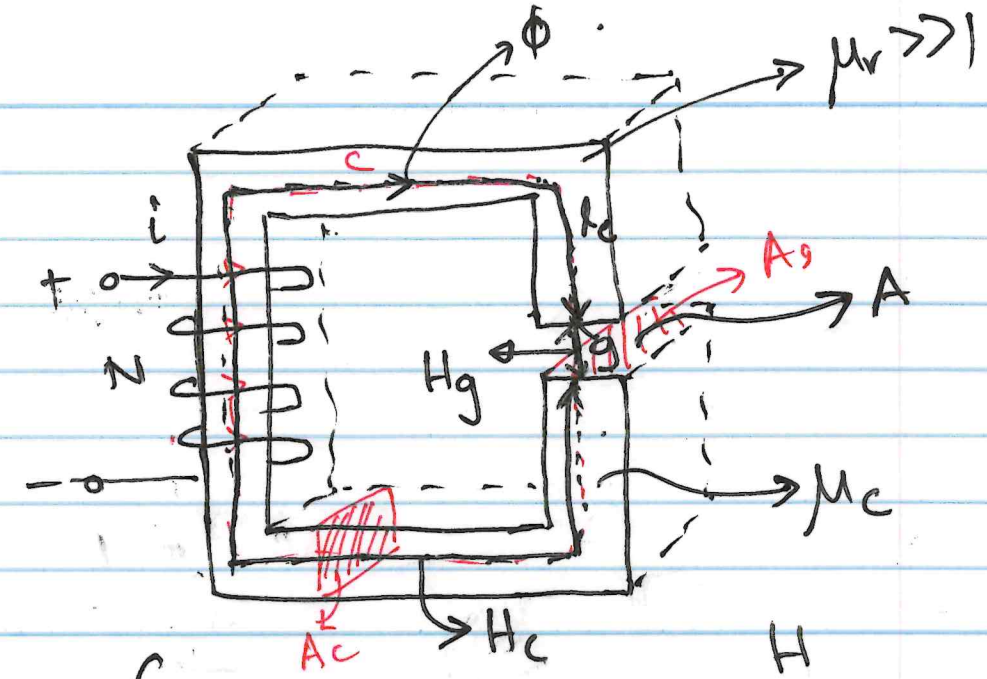
$$\oint_C \bar{H} \cdot d\bar{l} = \iint_S \left(\bar{J} + \frac{\partial \bar{D}}{\partial t} \right) \cdot d\bar{A}$$



$$\oint_C H \cdot dl = I$$

$$H \cdot \int_0^{2\pi} r \, d\theta = I$$

$$\boxed{H = \frac{NI}{2\pi r}}$$



$$\oint_c \bar{H} \cdot d\bar{l} = Ni$$

$$\Downarrow$$

$$H_c l_c + H_g g = Ni$$

$B = \mu H$ → $\mu_0 \mu_r$ → unit ka? → $4\pi \times 10^{-7}$ H/m

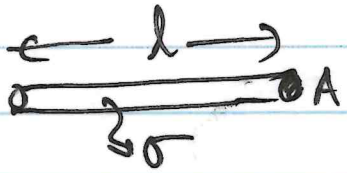
$$B_c = \mu_0 \mu_c H_c \qquad B_g = \mu_0 H_g$$

$$\phi_c = B_c A_c \qquad \phi_g = A_g B_g .$$

$$\phi_c = \mu_0 \mu_c H_c A_c \; ; \; \phi_g = \mu_0 A_g H_g .$$

$$\frac{\phi_c}{\mu_0 \mu_c A_c} \cdot l_c + \phi_g \cdot \frac{g}{\mu_0 A_g} = Ni$$

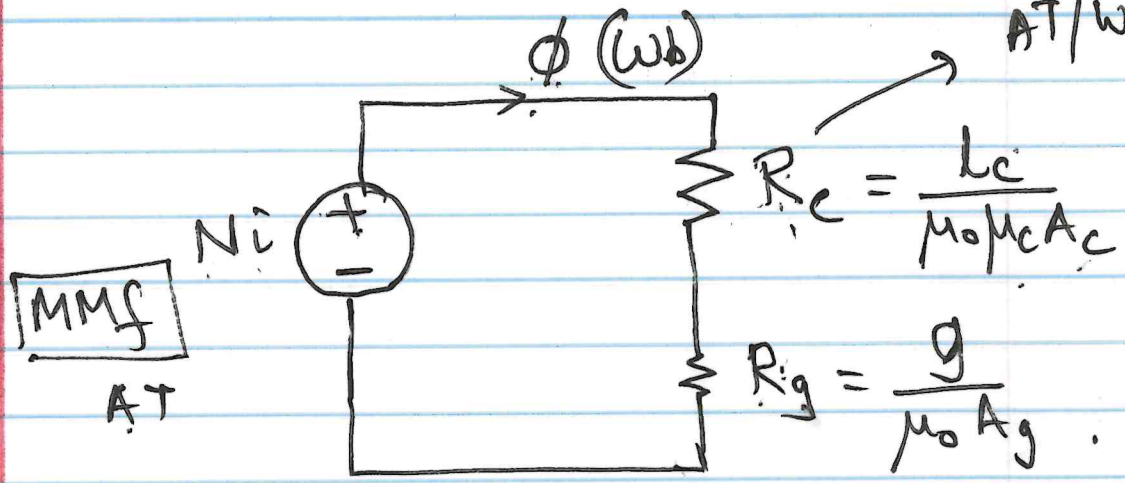
$$\phi \left[\frac{l_c}{\mu_0 \mu_c A_c} + \frac{g}{\mu_0 A_g} \right] = Ni$$



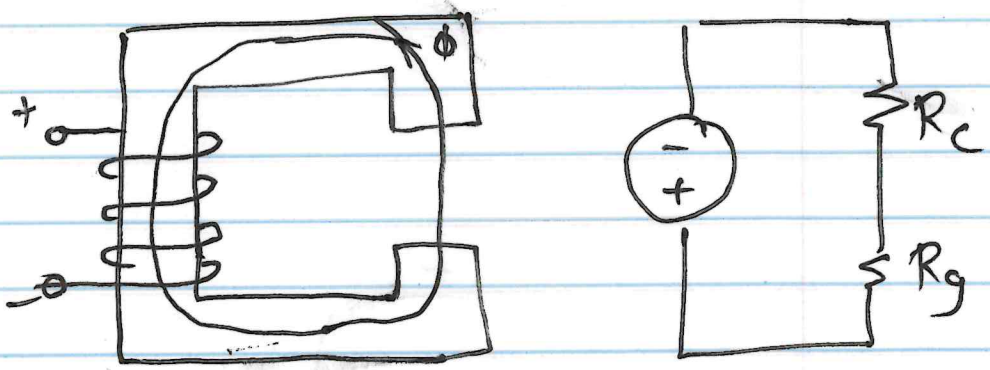
$$= R = \frac{l}{\sigma A}$$

$$\text{Reluctance} = \frac{l}{\mu_0 \mu_c A}$$

$$\phi \left[R_c + R_g \right] = Ni$$



Magnetic Equivalent circuit



$$v = L \frac{di}{dt}$$

$i \rightarrow H \rightarrow B \rightarrow \phi \rightarrow v$

$$\lambda = N\phi \Rightarrow v = \frac{d\lambda}{dt}$$

$$v = \frac{d\lambda}{dt}$$

$$= N \frac{d}{dt} \phi$$

$$= N \frac{d}{dt} \frac{Ni}{R_T} \qquad R_T = R_c + R_g$$

$$= \boxed{\frac{N^2}{R_T}} \frac{di}{dt}$$

